

MASS-APP-014-V12 — Trust, Reliability & Execution Assurance

Document Control

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1. Purpose

V12 proves whether accepted responsibilities were completed correctly, on time, and with visible evidence. It observes execution, detects failure, coordinates governed recovery, verifies outcomes, and returns concise assurance without becoming the workflow or operational system of record.

2. Permanent Architecture and Boundary

V12 owns responsibility-acceptance evidence, checkpoints, assurance cases, reliability observations, exceptions, reconciliation findings, trust indicators, and assurance briefings. ENG-006 executes workflows; ENG-011 observes infrastructure; ENG-013 governs errors; ENG-020 owns operations; ENG-021 owns dispatch; ENG-025 governs automation. V12 verifies their reported outcomes and never performs their work.

3. Platform Consumption Map

Capability | Consumption

ENG-003/004 | Principal identity and authorization

ENG-005 | Execution lifecycle events

ENG-006 | Workflow state and retry requests

ENG-007/008 | Knowledge and evidence references

ENG-009 | NOVA/POPS advisory requests

ENG-011/013 | Observability and governed failures

ENG-020/021/025 | Operations, mission, and automation outcomes

ENG-024 | Reliability analysis and trends

APP-014 V03/V09/V11 | Decision evidence, execution intent, executive awareness

Publishes assurance.checkpoint.failed, assurance.outcome.verified, execution-exception.raised, reconciliation.completed, and reliability-score.updated.

4. Role Mapping

V12 role | Baseline mapping | Authority

Assurance Viewer | Viewer specialization | View authorized assurance state

Responsibility Owner | Contributor specialization | Accept responsibility and submit evidence

Assurance Steward | Steward specialization | Define checkpoints and verify outcomes

Assurance Authority | Administrator specialization | Approve policy, escalation, and closure

The roles specialize APP-014 baseline roles; they do not replace them. The owner cannot verify their own final outcome when independent verification is required.

5. Capability Model

Responsibility Acceptance

An acceptance record binds a responsible principal, authoritative work reference, commitment, due time, service expectation, and required evidence. It does not create the underlying task.

Checkpoints and Evidence

Checkpoints define observable completion conditions. Evidence remains in ENG-008 or the authoritative source; V12 stores immutable references, hashes, submitter, observation time, and validation result.

Monitoring and Failure Detection

Deadlines, SLA thresholds, missing events, invalid evidence, and contradictory source states create findings. Detection does not itself retry or mutate execution.

Escalation, Retry, and Recovery

V12 recommends or requests recovery through ENG-006/ENG-025. Retry requests are idempotent and policy-bound. Human approval remains mandatory where risk, expenditure, external communication, or constitutional authority requires it.

Reconciliation and Outcome Validation

Reconciliation compares expected and observed states across governed sources. Outcome verification records checkpoint results, unresolved conflicts, verifier identity, and final disposition.

Reliability and Trust Indicators

Reliability scores summarize verified completion, timeliness, evidence sufficiency, recovery behavior, and reconciliation exceptions. Scores are explainable observations, not identity, relationship, employment, or punitive decisions.

NOVA and POPS

NOVA assurance analysis and POPS reliability history are advisory request types executed through ENG-009. They cannot close an assurance case or approve an exception.

6. Lifecycles

Responsibility: Proposed → Accepted → In Progress → Submitted → Verified → Closed

Exception: Detected → Assigned → Investigating → Recovery Requested → Resolved → Verified → Closed
Assurance case: Open → Collecting Evidence → Ready for Verification → Assured / Not Assured → Archived

Verified evidence and closed assurance cases are immutable. Corrections append superseding records.

7. Data Model

Entity | Purpose

AssuranceCase | Governed verification container

ResponsibilityAcceptance | Accepted obligation

VerificationCheckpoint | Required completion condition

CompletionEvidence | Immutable evidence reference

CheckpointResult | Verification outcome

ExecutionException | Detected failure or discrepancy

RecoveryRequest | Idempotent governed recovery request

ReconciliationRun | Cross-system comparison

ReconciliationFinding | Expected-versus-observed difference

ReliabilityObservation | Explainable reliability input

TrustIndicator | Aggregated assurance signal

AssuranceBriefing | Executive assurance summary

AdvisoryRequest | NOVA or POPS advisory request

All relational references are tenant-safe. Source references are read-only and validated against tenant, lifecycle, and authorization at creation and retrieval.

8. API Contracts

Method | Path | Purpose

POST/GET | /assurance-cases | Create or browse cases

GET | /assurance-cases/{id} | Read case and status

POST | /responsibilities/{id}/accept | Accept responsibility

POST | /assurance-cases/{id}/evidence | Submit evidence reference

POST | /checkpoints/{id}/verify | Record verification

GET | /execution-exceptions | Browse exceptions

POST | /execution-exceptions/{id}/assign | Assign exception owner

POST | /execution-exceptions/{id}/recovery | Request governed recovery

POST | /reconciliations | Run reconciliation

GET | /reconciliations/{id}/findings | Read findings

GET | /reliability/{subjectType}/{subjectId} | Explain reliability

POST | /assurance-cases/{id}/close | Close verified case

GET | /assurance-briefings | Browse briefings

POST | /assurance-advisory | Request NOVA/POPS advisory

9. Security and Database Integrity

Concrete RLS restricts every tenant-owned record. Composite foreign keys prevent cross-tenant relationships. Database triggers prevent mutation of verified evidence, verified checkpoint results, and closed cases. A verification trigger blocks required independent verification by the responsibility owner. Recovery requests carry unique idempotency keys.

Implementation-grade definitions are in MASS-APP-014-V12_Migration_Reference.sql.

10. V1 Implementation

V1 supports responsibility acceptance, checkpoint monitoring, evidence references, deadline and SLA detection, exception assignment, governed retry requests, reconciliation, explainable reliability scoring, assurance briefings, and human verification.

11. Future Evolution

Additional source adapters, richer statistical reliability models, and predictive exception awareness may attach through existing observation and advisory contracts. No future capability may silently convert an assurance signal into autonomous discipline or execution.

12. Constitutional Boundary Statement

V12 verifies execution; it does not own execution. It does not create operational truth, replace domain owners, mutate source systems, determine identity or relationships, or remove human accountability.